

| Question |     |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks available |     |     |       |       |      |
|----------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) |      | This is the energy needed to eject an electron (1)<br>from the surface <b>or</b> the metal <b>or</b> rubidium ( <b>not</b> the atom) (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2               |     |     | 2     |       |      |
|          | (b) |      | Any electron ejected is ejected by a single photon or equivalent (1)<br>Photon energy [or $hf$ ] must be [at least] equal to $\phi$ (1)<br>So $hf > \phi$ [for emission] (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1<br>1          | 1   |     | 3     |       |      |
|          | (c) | (i)  | Mean = 0.377 [V] Accept 0.376 [V] <b>or</b> 0.3765 [V] <b>or</b> 0.38 (V) (1)<br>$\pm 0.011$ [V] <b>or</b> 0.0110 [V] <b>or</b> 0.01 [V] (1)<br>Answer + uncertainty expressed to an appropriate number of sig figs i.e. 0.377 (or 0.376) V $\pm$ 0.011 V or 0.38 $\pm$ 0.01 [V] (1) Must use earlier calculations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 | 3   |     |       | 3     | 3    |
|          |     | (ii) | $E_{k \max}$ calculated for <b>one</b> stopping pd i.e. $0.366 \text{ V} \rightarrow 5.86 \times 10^{-20} \text{ J}$ ,<br><b>or</b> $0.388 \text{ V} \rightarrow 6.21 \times 10^{-20} \text{ J}$ <b>or</b> ( $0.377 \text{ V} \rightarrow 6.03 \times 10^{-20} \text{ J}$ ) (1)<br>Einstein's equation used correctly, either with $hf$ and <b>one</b> $E_{k \max}$<br>value including mean to give $\phi$ , <b>or</b> with $hf$ and $\phi = 3.62 \times 10^{-19} \text{ J}$<br>to give $E_{k \max}$ e.g.<br><br>$\phi = 6.63 \times 10^{-34} \times 6.34 \times 10^{14} - 5.86 \times 10^{-20} \text{ (1)}$ $= 3.62 \times 10^{-19} \text{ [J] (1)}$<br><b>or</b> $\phi = 6.63 \times 10^{-34} \times 6.34 \times 10^{14} - 6.21 \times 10^{-20} \text{ (1)}$ $= 3.58 \times 10^{-19} \text{ [J] (1)}$<br><b>or</b> $E_{k \max} = 6.63 \times 10^{-34} \times 6.34 \times 10^{14} - 3.62 \times 10^{-19} \text{ (1)}$ $= 5.83 \times 10^{-20} \text{ [J] (1)}$ |                 |     |     |       |       |      |

| Question |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available |          |          |           |          |          |
|----------|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |  |  | Conclusion: the $\phi$ of Rb, ( $3.62 \times 10^{-19}$ J ) just fits in the range $3.58 \times 10^{-19}$ J to $3.62 \times 10^{-19}$ J so Rb is a possibility <b>or</b> $E_{k\max} = 5.83 \times 10^{-20}$ J and lies just outside the range $5.86 \times 10^{-20}$ J to $6.21 \times 10^{-20}$ J) so possibly Rb or can't be Rb (1)<br><br><b>Variation on this method for the 1<sup>st</sup> and 4<sup>th</sup> marks:</b><br>$5.83 \times 10^{-20}$ J correspond to a stopping pd of 0.364 V (1)<br>which lies just outside the range given, so possibly / probably not Rb (1) |                 |          | 4        | 4         | 2        | 4        |
|          |  |  | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>4</b>        | <b>4</b> | <b>4</b> | <b>12</b> | <b>5</b> | <b>7</b> |